IT Professional From Scratch

Lessons 1–4 • Beginner Foundations

Learning approach: Learn → Practice → Troubleshoot → Build → Review

This guide collects the first four lessons of your IT learning path into one study document. Work through each lesson, perform the practical labs, and keep your command-line results for review.

Roadmap

Stage	Focus
1	Computer fundamentals, Windows, hardware and troubleshooting
2	Networking fundamentals
3	Linux and system administration
4	Python, Git/GitHub and automation
5	SQL, databases, web and APIs
6	Cybersecurity fundamentals
7	Cloud, Docker and DevOps
8	Professional IT skills, projects and portfolio

Lesson 1 — What Is IT?

IT means Information Technology. At its simplest, IT is the use and management of computers, software, networks, data and technology systems to solve problems.

What an IT professional may manage

• Computers and operating systems
• Networks and Wi-Fi
• User accounts and permissions
• Software and applications
• Servers and databases
• Security and backups
• Websites, APIs and cloud systems
• Troubleshooting and technical support

Core mental model

```
User
  ↓
Input
  ↓
Application
  ↓
Operating System
  ↓
Hardware / Network
  ↓
Output
```

> The goal is not simply to know how to use technology. You will learn how technology works, how to diagnose failures, how to secure systems and how to build useful solutions.

Starter practical exercise

On Windows press Windows + R, type **msinfo32**, and inspect Processor, Installed Physical Memory (RAM), System Type and Windows Version.

Then open Command Prompt and run:

```
hostname
whoami
ipconfig
```

Lesson 2 — How a Computer Actually Works

A computer receives input, processes information, stores information and produces output. The major pieces work together rather than operating independently.

CPU — Central Processing Unit

The CPU executes instructions. It contains processing cores and operates at a clock frequency, but CPU performance depends on architecture, generation, cache, cores and workload—not clock speed alone.

RAM — Random Access Memory

RAM is temporary working memory used by the operating system and running applications. A useful analogy is a desk: the SSD is the bookshelf, RAM is the desk and the CPU is the brain doing the work.

Storage — SSD / HDD

Storage keeps programs and files when the computer is powered off. HDDs use spinning magnetic media; SSDs use flash memory and are generally much faster and quieter.

GPU — Graphics Processing Unit

The GPU specializes in graphics and highly parallel workloads such as gaming, 3D rendering, video processing and some AI workloads.

Motherboard

The motherboard connects major components such as the CPU, RAM, storage, GPU and peripheral interfaces.

Network adapter

Wi-Fi and Ethernet adapters allow the computer to communicate with local networks and the internet.

Operating system and drivers

The operating system coordinates hardware and provides services to applications. Drivers allow Windows to communicate with specific hardware devices.

Example: opening Chrome

```
Mouse click
  ↓
Windows receives input
  ↓
Windows locates Chrome on SSD
  ↓
Required data is loaded into RAM
  ↓
CPU executes instructions
  ↓
GPU helps render graphics
  ↓
Display shows Chrome
```

Practical lab: open Task Manager with Ctrl + Shift + Esc, select Performance, and inspect CPU, Memory, Disk, Wi-Fi/Ethernet and GPU.

Lesson 3 — How Windows Manages Your Computer

Windows manages running programs, memory, files, users, permissions, devices and background services.

Programs vs processes

A program is software stored on disk. A process is a running instance of a program. When you launch an application, Windows creates one or more processes and gives them access to system resources.

Task Manager

Task Manager lets you observe processes and resource usage. CPU shows processing activity; Memory shows RAM usage; Disk shows storage activity; network and GPU sections show other resource activity.

Background processes and services

Windows and installed software run background processes and services for networking, updates, security, printing and other functions. Do not terminate or disable an unfamiliar process/service without understanding its purpose.

Files, folders and paths

```
C:\Users\YourName\Downloads
```

A path tells the operating system where an item is located. File extensions such as .pdf, .jpg, .mp3, .exe, .py and .html commonly indicate file types.

Users and permissions

Windows supports different user accounts and permissions. Users may have rights to read, modify, delete, execute or administer resources. Understanding permissions is essential for system administration and cybersecurity.

Command-line practice

```
hostname
whoami
cd Desktop
dir
cd ..
cd Downloads
dir
```

Troubleshooting method

Observe → Identify → Hypothesize → Test → Fix → Verify

Example: if a computer is slow, inspect CPU, Memory and Disk usage first. Identify unusually heavy processes, form a hypothesis, make a controlled change, then verify whether the problem improved.

Lesson 4 — Networking Fundamentals

Networking is how computers and other devices communicate. It is foundational for IT support, system administration, cybersecurity, cloud and software development.

Network and LAN

A network is a group of connected devices that communicate and share resources. A LAN (Local Area Network) normally covers a home, office, building or similar local area.

Internet and ISP

The internet is a global network of interconnected networks. Your ISP (Internet Service Provider) connects your local network to the wider internet.

IP addresses

An IP address identifies a network interface at the IP layer. A typical private IPv4 address might look like 192.168.1.25. Private IPv4 ranges include 10.0.0.0/8, 172.16.0.0/12 and 192.168.0.0/16.

Router and default gateway

A router forwards traffic between networks. Your default gateway is normally the local router used when traffic needs to reach destinations outside the local subnet.

DHCP

DHCP (Dynamic Host Configuration Protocol) can automatically provide devices with an IP address, subnet configuration, default gateway and DNS server information.

DNS

DNS (Domain Name System) maps domain names to IP addresses and supports many other DNS record types. It allows people to use names such as google.com instead of remembering IP addresses.

Ports

A port helps identify a network service endpoint. Common conventional ports include 22 for SSH, 53 for DNS, 80 for HTTP, 443 for HTTPS, 3306 for MySQL and 5432 for PostgreSQL.

HTTP and HTTPS

HTTPS is HTTP protected with TLS. It helps provide confidentiality and integrity for traffic between a client and server when TLS is correctly configured.

Packets, TCP and UDP

Network communication is commonly carried in packets. TCP provides reliable, ordered delivery with mechanisms such as retransmission and congestion control. UDP is simpler and connectionless, with less protocol overhead and without TCP's built-in reliability mechanisms.

MAC addresses

A MAC address is associated with a network interface and operates at the data-link layer. Do not confuse a MAC address with an IP address; they operate at different layers and have different roles.

OSI model

```
7  Application
6  Presentation
5  Session
4  Transport
3  Network
2  Data Link
1  Physical
```

First networking lab

```
ipconfig
ping 127.0.0.1
ping YOUR_DEFAULT_GATEWAY
nslookup google.com
ping google.com
```

Interpretation: 127.0.0.1 tests the local IPv4 loopback. Your gateway test checks local network reachability. nslookup tests DNS resolution. ping to a domain tests reachability using ICMP when the destination/network permits it.

Lessons 1–4 Review

Key concepts to know before continuing

Concept	Can you explain it?
CPU	What does it do? What are cores and clock speed?
RAM	Why is it different from storage?
SSD/HDD	What is the difference?
Operating system	Why does Windows sit between applications and hardware?
Driver	Why can a driver affect hardware functionality?
Process	How is it different from a program?
Service	Why do background services exist?
IP address	What is it used for?
Router	What does it do?
DHCP	Why is automatic network configuration useful?
DNS	Why do we need it?
Port	How is it different from an IP address?
TCP/UDP	What is the main difference?
MAC address	What layer is it associated with?

Professional mindset

Do not guess first. Observe the system, collect evidence, form a hypothesis, test one controlled change, and verify the result. This mindset will be used throughout networking, Linux, cybersecurity, programming and cloud administration.

Next lesson: IP Addresses and Subnetting

You will learn subnet masks, CIDR notation such as /24, network and broadcast addresses, usable host ranges and how to determine whether two devices are on the same subnet.